NATIONAL HURRICANE CENTER

TROPICAL CYCLONE FORECAST VERIFICATION

1973

following tables contain verification statistics for the 1973 tropical cyclone forecasts. The first table contains error summaries for the National Hurricane Center advisory forecasts (designated OFFICIAL) together with those for all operationally available objective techniques. The second and third tables contain homogeneous samples of various groupings of the OFFICIAL forecasts with different objective techniques. The fourth table is a comparison of the preliminary N.H.C. and N.M.C. forecasts and the final (coordinated) forecasts.

The forecast errors, expressed in terms of nautical miles, are computed in the following way. The magnitude of the vector difference (along a great circle) between the forecast displacement for a given time period and the actual displacement taken from the 'best track' is computed for each forecast. All forecasts for which a given cyclone maintains tropical storm of hurricane intensity are verified. These results are then averaged and tabulated.

The errors in the initial positioning of storms are also computed. For a given forecast, this is simply the magnitude of the great circle distance between the operational initial position and the corresponding 'best track' initial position.

In the N.H.C.-N.M.C. comparison (fourth table) no account was taken of the initial positioning errors. In this table a forecast error simply represents the distance between a corresponding forecast position and 'best track' position

1973 TROPICAL CYCLONE FORECAST VERIFICATION

HOMOGENEOUS SAMPLE OF N.H.C. AND N.M.C. PRELIMINARY FORECASTS AND OFFICIAL FORECASTS

FORECAST TYPE	FORECAST PERIOD			
	12 HOUR	24 HOUR	48 HOUR	72 HOUR
N.H.C. PRELIMINARY	52	106	235	401
N.M.C. PRELIMINARY	59	126	298	440
OFFICIAL FORECASTS	51	107	239	421
(number of cases)	(47)	(41)	(25)	(14)

(Forecast errors are expressed in terms of vector errors in nautical miles.)

1973 TROPICAL CYCLONE FORECAST VERIFICATION

FORECAST TYPE (no. of cases)	FORECAST PERIOD				
	INITIAL POSITION	12 HOUR	24 HOUR	48 HOUR	72 HOUR
OFFICIAL	24 (98)	57 (98)	107 (84)	239 (54)	346 (28)
HURRAN	24 (60)	56 (60)	122 (54)	293 (40)	383 (20)
CLIPER	22 (95)	59 (95)	131 (83)	299 (55)	385 (27)
SANBAR	16 (43)	55 (43)	103 (37)	295 (25)	480 (12)
NHC-67	18 (86)	50 (86)	91 (74)	228 (50)	322 (22)
NHC-72	22 (95)	56 (95)	114 (83)	269 (55)	350 (27)
NHC-73	17 (43)	45 (43)	87 (37)	226 (25)	353 (11)

DISPLACEMENT ERRORS IN NAUTICAL MILES

1973 TROPICAL CYCLONE FORECAST VERIFICATION

HOMOGENOUS SAMPLE

FORECAST TYPE (no. of cases)	FORECAST PERIOD				
	INITIAL POSITION	12 HOUR	24 HOUR	48 HOUR	72 HOUR
OFFICIAL	18 (86)	52 (86)	102 (74)	249 (48)	397 (22)
CLIPER		54	116	290	370
NHC-67		50	91	226	322
NHC-72		51	96	274	386

DISPLACEMENT ERRORS IN NAUTICAL MILES

1973 TROPICAL CYCLONE FORECAST VERIFICATION

HOMOGENEOUS SAMPLES

FORECAST TYPE (no. of cases)	FORECAST PERIOD I.P.	12 HOUR	24 HOUR	48 HOUR	72 HOUR
OFFICIAL	22 (60)	55 (60)	107 (54)	251 (38)	324 (20)
HURRAN		58	125	300	384
OFFICIAL	17 (43)	54 (43)	97 (37)	248 (24)	387 (12)
SANBAR		55	103	290	480
OFFICIAL	17 (43)	54 (43)	99 (37)	251 (24)	392 (11)
NHC-73		48	95	238	396

DISPLACEMENT ERRORS IN NAUTICAL MILES

VERIFICATION OF FORECASTS

OFFICIAL

POSITION ERROR (N.M.)	DISPLACEMENT ERROR		
	24 HOURS	48 HOURS	72 HOURS
17 (56)	76 (34)	187 (12)	356 (2)
23 (197)	101 (176)	238 (133)	385 (113)
1972 21 (63)	135 (55)	365 (36)	688 (23)
22 (316)	105 (265)	259 (181)	435 (138)

HURRAN

1970	83 (25)	160 (11	337 (2)
	111 (83)	279 (48)	444 (37)
1972	140 (22)	399 (17)	510 (15)
MEAN	112 (130)	288 76)	458 (54)

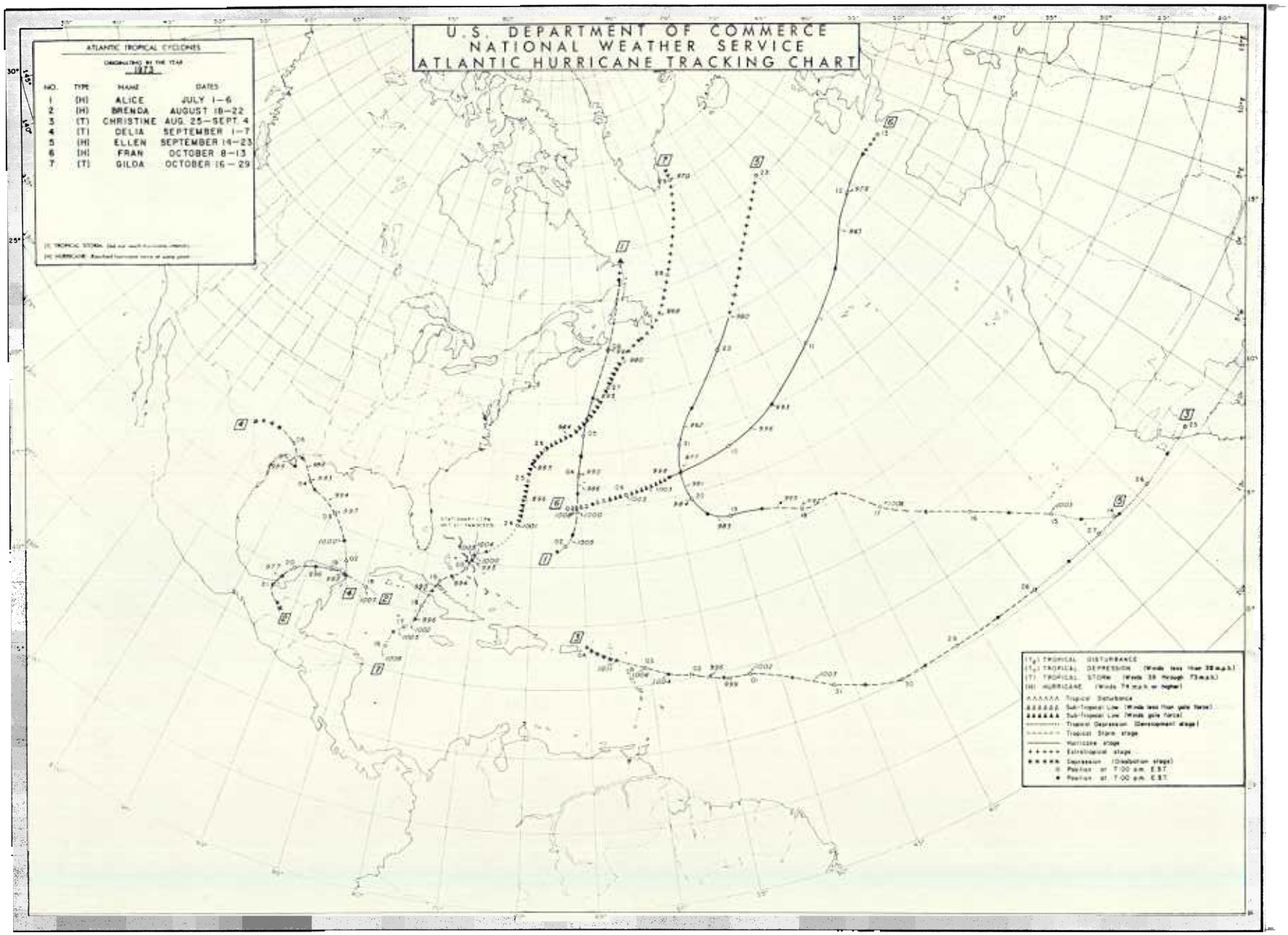
NHC-67

	92 (30)	260 (13)	480 (2)
	108 (176)	312 (136)	551 (115)
1972	117 (22)	301 (16)	431 (10)
	114 (246)	333 (177)	591 (138)

SANBAR

	138 (11)	218 (3)	431 (1)
	147 (81)	252 (64)	357 (54)
1972	117 (22)	301 (16)	431 (10)
	140 (114)	260 (83)	370 (65)

(Number in parentheses are the number of forecasts)


U.S. DEPARTMENT OF COMMERCE
NATIONAL WEATHER SERVICE
ATLANTIC HURRICANE TRACKING CHART
ATLANTIC TROPICAL CYCLONES
DESIGNATED IN THE YEAR 1973
NO. TYPE NAME DATES
1 (H) ALICE JULY 1–6
2 (H) BRENDA AUGUST 18–22
3 (T) CHRISTINE AUG. 25–SEPT. 4
4 (T) DELIA SEPTEMBER 1–7
5 (H) ELLEN SEPTEMBER 14–23
6 (H) FRAN OCTOBER 8–13
7 (T) GILDA OCTOBER 16–29
(T) TROPICAL STORM (Winds 39 through 73 m.p.h.)
(H) HURRICANE (Winds 74 m.p.h. or higher)
Tropical Disturbance
Sub-Tropical Low (Winds gale force)
Tropical Depression (Development stage)
Tropical Storm stage
Hurricane stage
Extratropical stage
Position at 7:00 p.m. E.S.T.

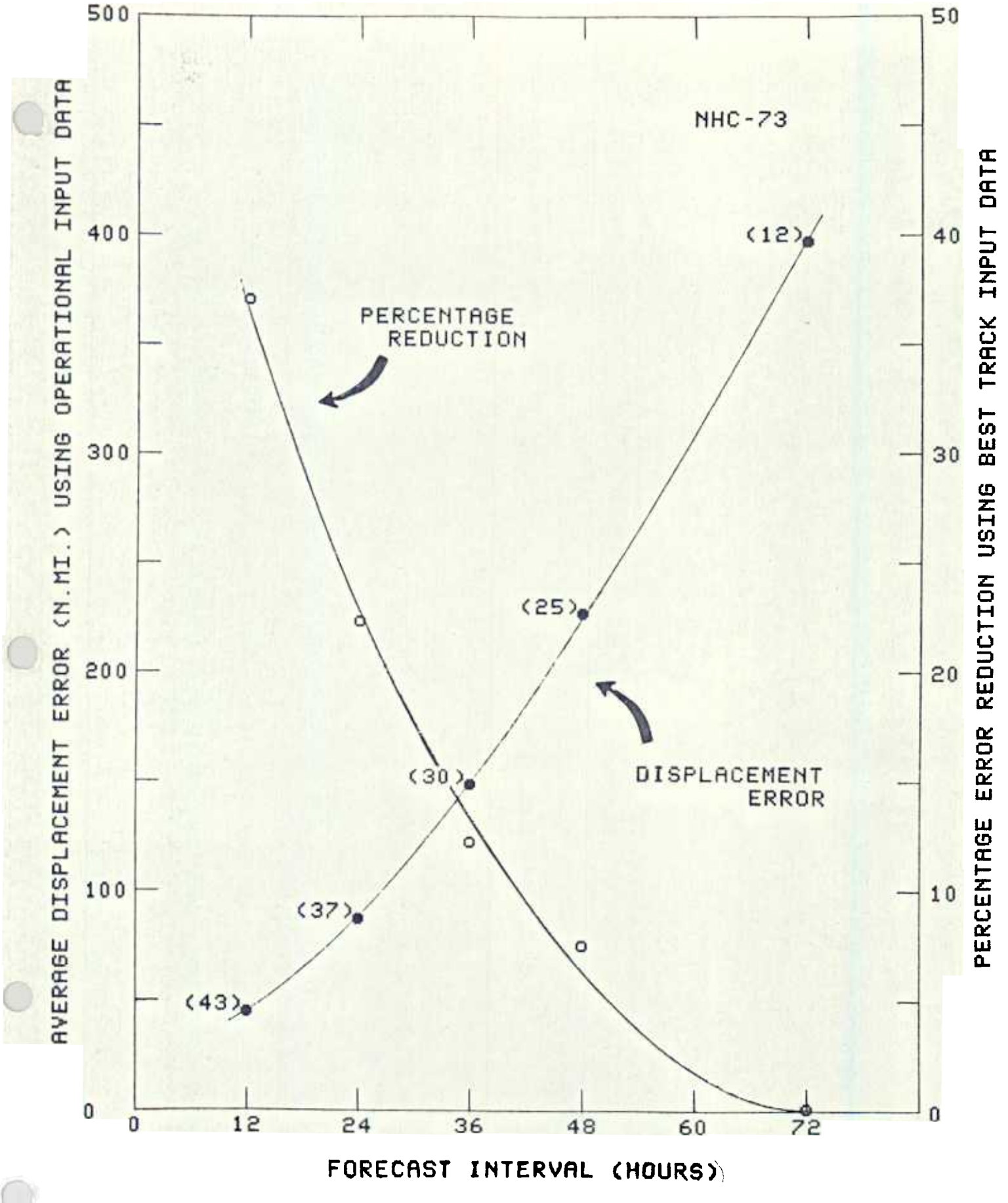
NMC's

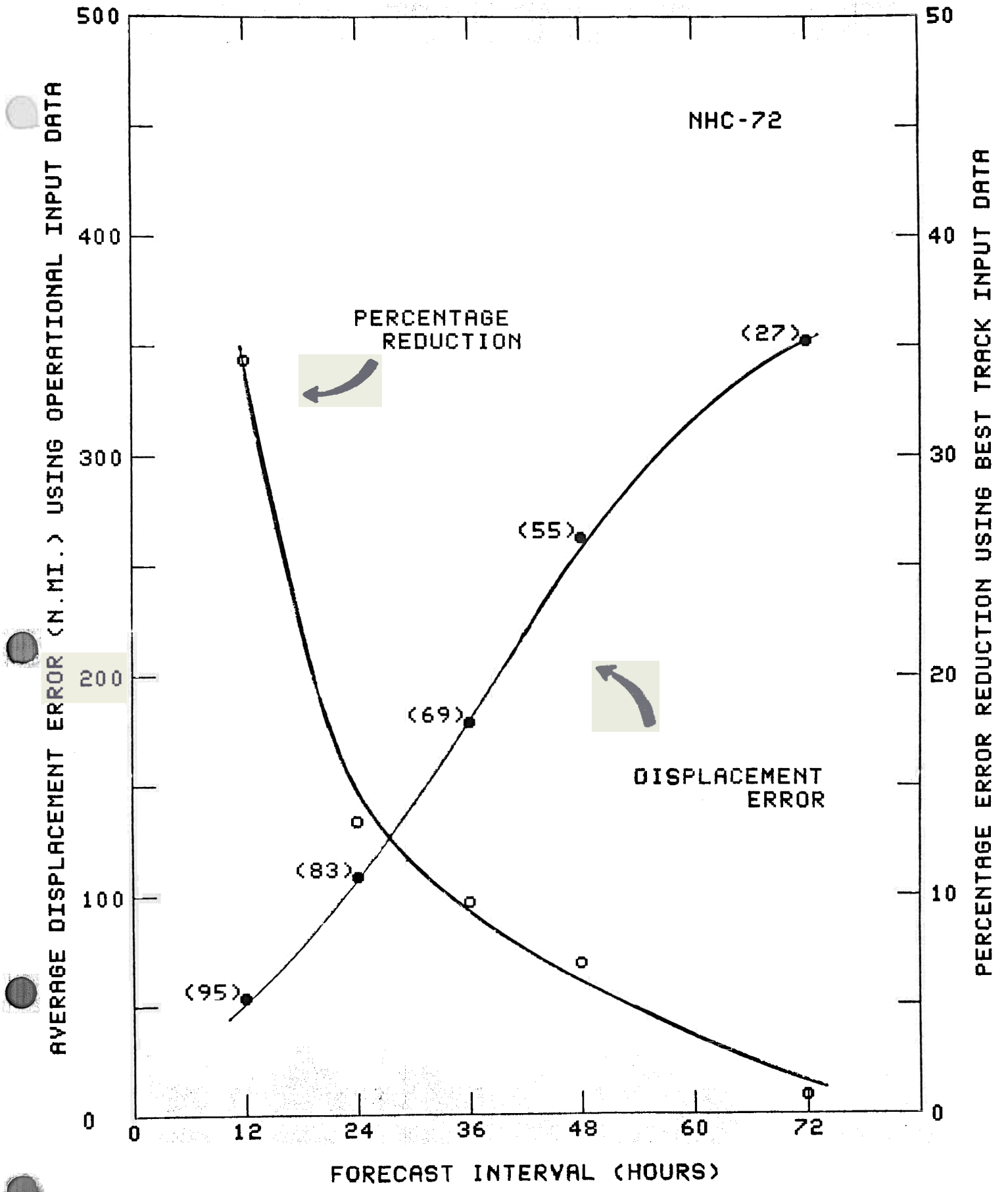
1973 TROPICAL CYCLONE FORECAST VERIFICATION

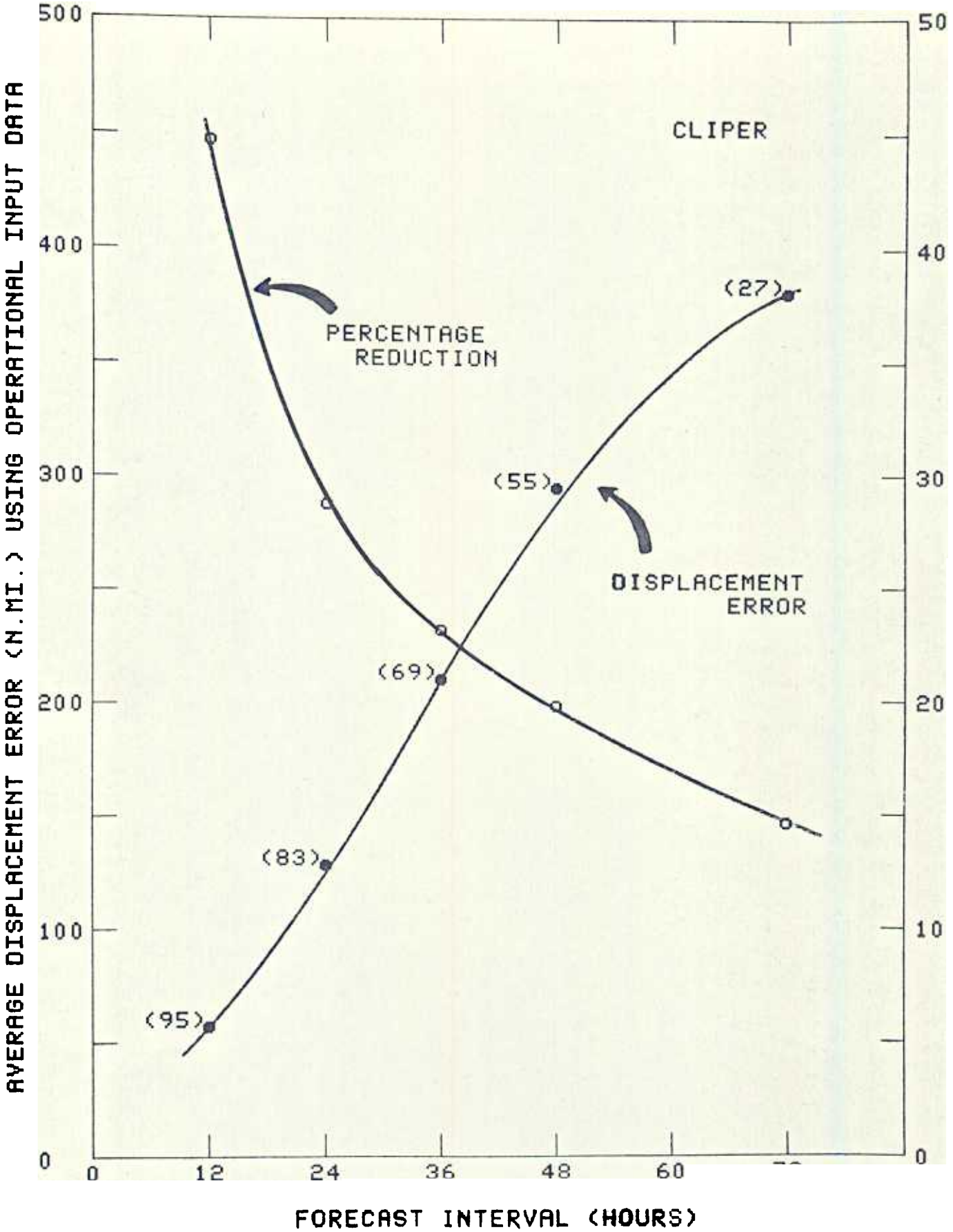
STORM (cases)	NHC			NMC		
	24HR	48HR	72HR	24HR	48HR	72HR
ALICE	66	144	378	108	234	528
	5	5	4	5	5	4
BRENDA	114	318	612	102	288	492
	11	7	3	11	7	3
DELIA	174	270	642	192	396	750
	17	11	5	17	11	5
GILDA	96	222	360	126	306	444
(to 232200)	24	19	14	24	19	14
ALL	120	241	446	139	318	521
	57	42	26	57	42	26

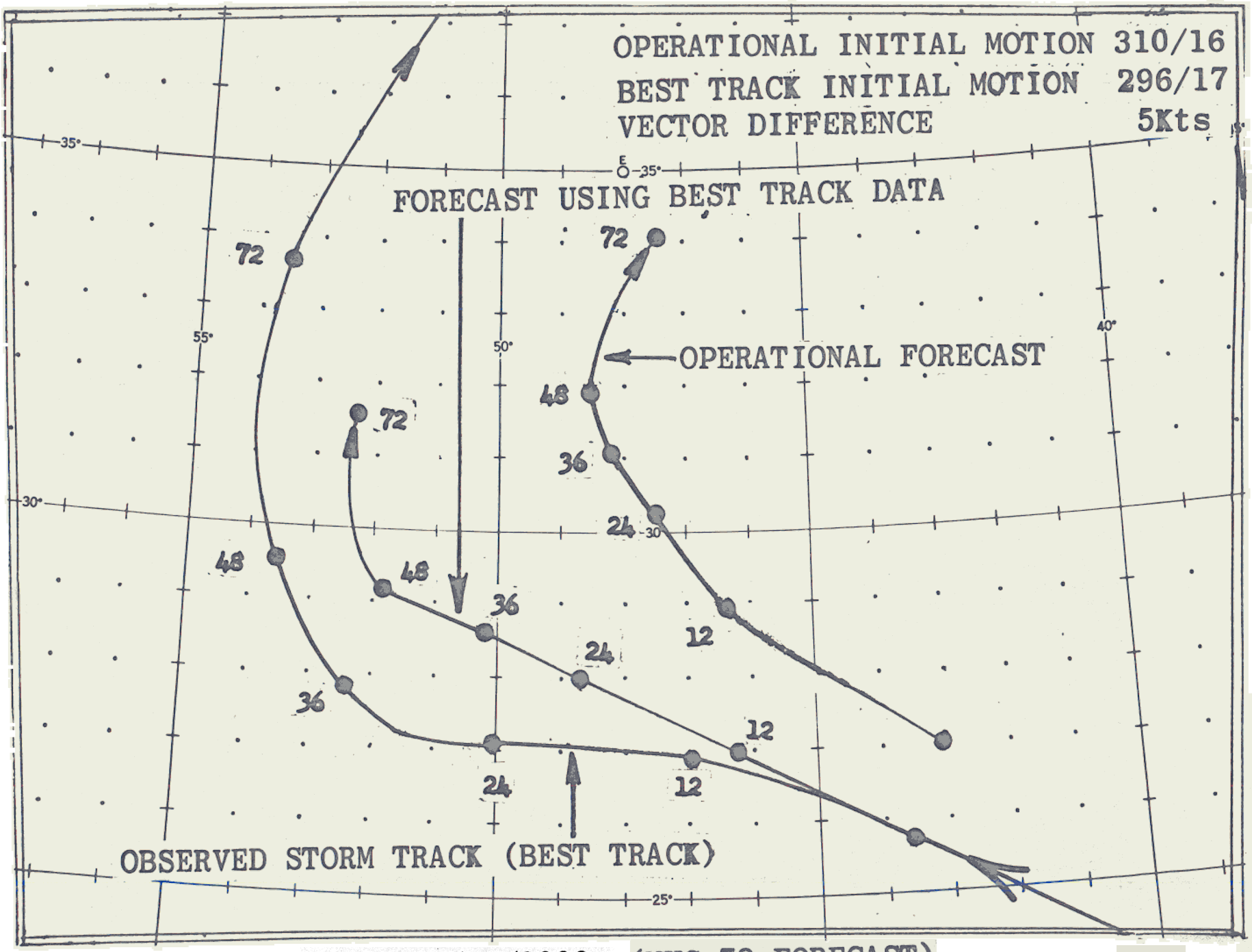
STORM (cases)	NHC			NMC		
	24HR	48HR	72HR	24HR	48HR	72HR
GILDA	116	276	480	116	234	456
(after 232200)	10	10	10	10	9	9

TABLE 3.

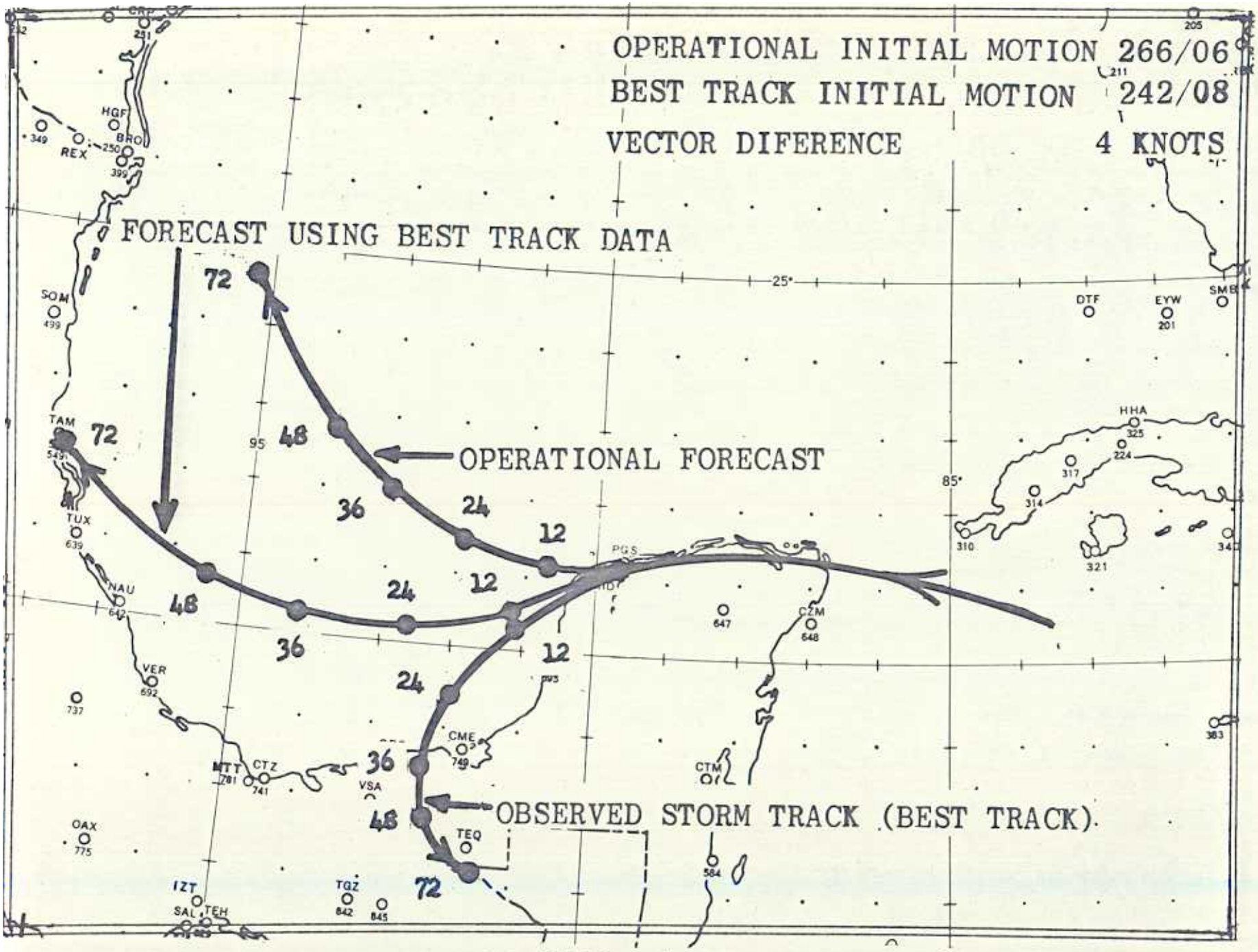

NHC-73
AVERAGE DISPLACEMENT ERROR (N.MI.) USING OPERATIONAL INPUT DATA
PERCENTAGE ERROR REDUCTION USING BEST TRACK INPUT DATA
FORECAST INTERVAL (HOURS)
PERCENTAGE REDUCTION
DISPLACEMENT ERROR
(12)
(25)
(30)
(37)
(43)
0
100
200
300
400
500
0
10
20
30
40
50
0
12
24
36
48
60
72


NHC-72
PERCENTAGE REDUCTION
DISPLACEMENT ERROR
(27)
(55)
(69)
(83)
(95)
AVERAGE DISPLACEMENT ERROR (N.MI.) USING OPERATIONAL INPUT DATA
PERCENTAGE ERROR REDUCTION USING BEST TRACK INPUT DATA
FORECAST INTERVAL (HOURS)
0
100
200
300
400
500
10
20
30
40
50
12
24
36
48
60
72

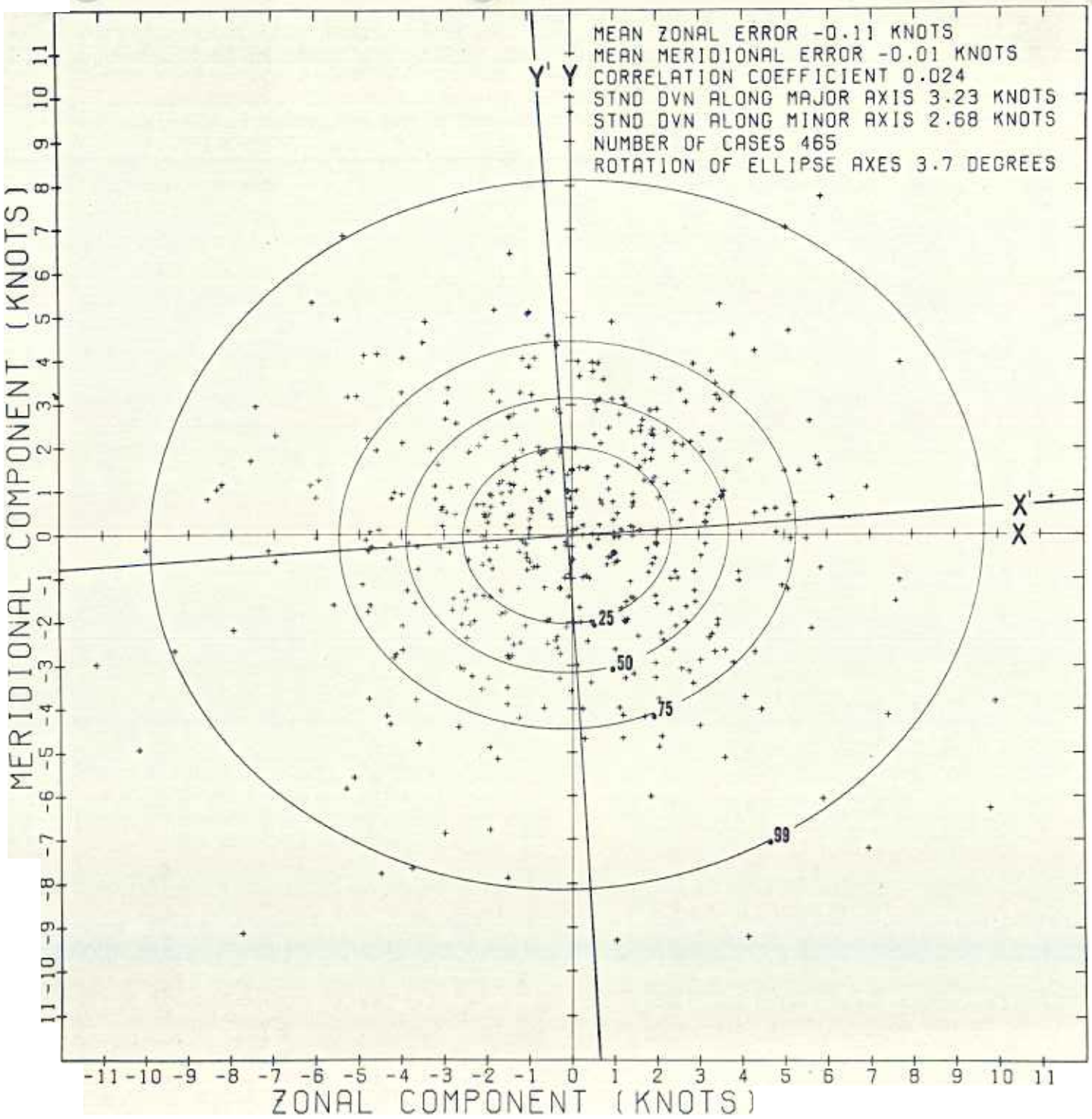

AVERAGE DISPLACEMENT ERROR (N.MI.) USING OPERATIONAL INPUT DATA
PERCENTAGE ERROR REDUCTION USING BEST TRACK INPUT DATA
FORECAST INTERVAL (HOURS)
CLIPER
PERCENTAGE REDUCTION
DISPLACEMENT ERROR
(27)
(55)
(69)
(83)
(95)
500
400
300
200
100
0
50
40
30
20
10
0
0
12
24
36
48
60



ELLEN 09/20/0000Z (NHC-72 FORECAST)



BRENDA 08/20/0000z (CLIPER FORECAST)


MEAN ZONAL ERROR -0.11 KNOTS
MEAN MERIDIONAL ERROR -0.01 KNOTS
CORRELATION COEFFICIENT 0.024
STND DVN ALONG MAJOR AXIS 3.23 KNOTS
STND DVN ALONG MINOR AXIS 2.68 KNOTS
NUMBER OF CASES 465
ROTATION OF ELLIPSE AXES 3.7 DEGREES
Y' Y
X' X
.25
.50
.75
.99
MERIDIONAL COMPONENT (KNOTS)
ZONAL COMPONENT (KNOTS)
-11 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11